

FIRST COURSE HANDOUT
A FIRST COURSE IN LINEAR ALGEBRA (MTH 201A)
2025-2026-I

CREDITS: 3-1-0-0 [11]

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1. PRE-REQUISITES

Basic set theory and Linear Algebra (MTH113).

2. LECTURE, TUTORIAL AND VENUE

- **Lectures:** Mondays, Wednesdays & Fridays at 17:00–18:00 hrs. **Venue:** LHC L-13
- **Tutorials:** Tuesdays at 17:00–18:00 hrs. **Venue:** LHC L-10

3. OFFICE HOURS

- **By appointment:** Appointments can be made by sending an email to **chary@iitk.ac.in**.

4. COURSE CONTENTS

We aim to broadly discuss the following topics in this course:

- (1) Basics on fields, systems of linear equations, row equivalence of matrices, row echelon matrices.
- (2) Vector spaces, subspaces, linear span, linear dependence and independence, basis and dimension of finite-dimensional vector spaces, row rank and column rank of a matrix and their equality, sum of subspaces and the dimension formula, coordinates in a finite-dimensional vector space.
- (3) Cosets, quotient spaces, direct sums of subspaces, complementary subspaces.
- (4) Linear transformations, isomorphisms, first isomorphism theorem, kernel and image, rank–nullity theorem, correspondence theorem, second and third isomorphism theorems, matrix representation, space of linear transformations.
- (5) Dual spaces, dual basis, double dual and the canonical isomorphism (finite-dimensional case), annihilators and hyperplanes.
- (6) Diagonalization, minimal polynomial, decomposition into eigenspaces, equivalent criteria for diagonalizability, simultaneous diagonalization.
- (7) Block-diagonal matrices, generalized eigenvectors and eigenspaces, decomposition into generalized eigenspaces (when the minimal polynomial splits), nilpotent operators, Jordan strings and Jordan basis, existence and uniqueness of Jordan canonical form, additive Jordan decomposition.
- (8) Determinants and their basic properties, Cramer’s rule, extension to polynomial matrices, characteristic polynomial, Cayley–Hamilton theorem.
- (9) Inner product spaces, norm, Cauchy–Schwarz inequality, orthogonality, orthogonal projection, Gram–Schmidt process, orthogonal decomposition, linear functionals and hyperplanes, orthogonal transformations, isometries of $\mathbb{R}^n$, cross product, Sylvester’s criterion for positive definiteness.

- (10) Inner product spaces over $\mathbb{C}$, self-adjoint, normal and unitary operators, and their spectral theorems.

If time permits, we may also discuss the following additional topic in lectures or discussion sessions:

- (1) Tensor products of vector spaces: definition and universal property, commutativity and associativity, tensor products of linear maps and matrices, extension of scalars, behavior with respect to composition, duals, etc.

5. HOMEWORK/EXERCISES

- (1) Exercises mentioned in lectures should be solved independently. No solution submissions are required. Nevertheless, students are strongly encouraged to discuss and solve problems from the recommended books. Some of these problems may be addressed during tutorial sessions.
- (2) A few topics will be introduced through exercises and problems. **Do not miss them**, as they form an integral part of the course.

6. REFERENCES

The topics covered can be found in the following standard texts:

- (1) *Linear Algebra* by K. Hoffman and R. Kunze.
- (2) *Linear Algebra and Its Applications* by Gilbert Strang.
- (3) *Algebra* by Michael Artin.
- (4) *Linear Algebra Done Right* by Sheldon Axler.

7. Evaluation Policy

- (1) **Absolute grading** will be followed for the final evaluation.
- (2) Evaluation will be based on the following components:
 - (a) **End-Semester Examination:** Weightage of **40%**.
 - (b) **Mid-Semester Evaluation:**
 - Weightage of **30%**, depending on the number of quizzes/assignments.
 - Quizzes may carry up to **20%**.
 - (c) **Sincerity:** This component reflects the student's consistent and honest effort to learn the subject. Students are expected to study regularly, attend discussion sessions, and make genuine attempts at solving exercises. The instructor may randomly ask any student to answer questions based on the lectures and tutorials covered up to that point. If a student is unable to explain something and has not informed the instructor in advance about their difficulties, it will be treated as negligence. The instructor will make an overall assessment of each student's sincerity. This judgment is entirely at the discretion of the instructor. The weightage for this component is **10%**.
 - (d) **About Letter Grades** The grade A^* is reserved for outstanding performance in the course. I reserve the right to award this grade based on overall performance: including exams, participation in tutorial sessions, and active engagement during the course. All other grades will be determined solely based on exam scores.
- (3) **Zero Tolerance for Malpractice:** Any similarity in answers, even for a single question, may invite strict action. If the use of unfair means is brought to notice in any of the evaluation components, disciplinary action as recommended by the SSAC will be taken. Repeat offenders may be deregistered from the course.

The instructor reserves the right to revise the course policy if deemed necessary. Students will be notified of any changes.